

- Q.21 Define electroplating and give two examples of its applications. (CO3)
- Q.22 Mention any two advantages and two limitations of CNC machines. (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the construction and working of a lathe machine with a neat diagram. (CO2)
- Q.24 Describe the working principle of Electric Discharge Machining (EDM) and its applications. (CO4)
- Q.25 Explain the process of MIG and TIG welding with their differences. (CO2)

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Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which of the following is NOT a mechanical property of a material? (CO1)
- a) Hardness b) Ductility
c) Elasticity d) Yielding
- Q.2 What is the main purpose of cutting fluids in machining operations? (CO1)
- a) Improve surface finish
b) Reduce tool wear
c) Dissipate heat
d) All of the above
- Q.3 Which type of lathe is most suitable for heavy-duty operations? (CO2)
- a) Light duty lathe b) Medium duty lathe
c) Heavy duty lathe d) None of these

- Q.4 In milling, which of the following is a type of milling operation? (CO2)
- a) Step turning b) Boring
- c) Knurling d) Indexing
- Q.5 Which of the following is a type of welding defect? (CO2)
- a) Porosity b) Knurling
- c) Boring d) Broaching
- Q.6 Which modern machining process uses dielectric fluid? (CO4)
- a) EDM b) ECM
- c) AJM d) CNC

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 The process of coating steel with a layer of zinc to prevent rusting is called _____. (CO3)
- Q.8 In CNC programming, G-Codes are used for _____ functions. (CO4)
- Q.9 The machine tool used for producing cylindrical surfaces is called a _____. (CO2)

- Q.10 The process of joining two metals by melting a filler material is called _____. (CO2)
- Q.11 The tool used in drilling operations to create holes is called a _____. (CO2)
- Q.12 The process of surface finishing by using abrasive particles in a liquid medium is called _____. (Co3)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Define heat treatment and name any two heat treatment processes. (CO1)
- Q.14 What are the functions of lubricants in machining? (CO1)
- Q.15 List any three difference between gas welding and are welding. (CO2)
- Q.16 Explain the principle of centerless grinding. (CO3)
- Q.17 What is indexing in milling? Mention its types. (CO2)
- Q.18 What is the purpose of a reamer in drilling? (CO2)
- Q.19 Explain the working principle of a radial drilling machine. (CO2)
- Q.20 What are the different types of cutting fluids used in machining? (CO1)